

and with proper conversion, supply 5V to the circuit boards. Connecting the batteries judiciously can also yield multiple voltage outputs. By connecting two 6V batteries in different ways, we can get 12V and 6V. This system isn't nearly as fool proof as it seems, however. More than likely, the two batteries will not be discharged at the same rate. This causes extra current to be drawn from one to the other, and the batteries may not last as long as they might otherwise. If all of the subsystems in your robot use the same batteries, be sure to add sufficient filtering capacitors across the positive and negative power rails. The capacitors help soak up current spikes and noise, which are most often contributed by motors. Place the capacitors as near to the batteries and the noise source as possible. A good rule of thumb is 100 μF of capacitance for every 250 mA of current drawn by a motor during normal operation. Be certain the capacitors you use are overrated above the voltage by 50 to 75 % (e.g. use a 7.5 V rated capacitor for a 5 V circuit). An underrated capacitor will probably burn out or possibly develop a short circuit, which might "fry" your circuit. You should place smaller value capacitors, such as 0.01 to 0.1 μF , across the positive and negative power rails wherever power enters or exits a circuit board. As a general rule, you should add one of these decoupling capacitors at the power input pins of all ICs. Linear ICs, such as the 555 timer need decoupling capacitors, or the noise they generate through the power lines can ripple through to other circuits.

Voltage Regulation

Most circuits require a precise voltage or they may get damaged or act erratically. Generally, provide voltage regulation only to those components and circuit boards in your robot that require it. You can easily add a variety of different solidstate voltage regulators to your electronic circuits. They are easy to obtain, and you can choose from among several styles and output capacities. In this chapter, some of the different types will be described along with their operating characteristics.

Zener diode voltage regulation

A quick and relatively small method for providing regulated voltage is to use zener diodes. With a zener diode, current does not begin to flow through the load circuitry until the voltage exceeds a certain level (called the breakdown voltage). Voltage over this level is then "shunted" through the zener diode, effectively limiting the voltage to the rest of the circuit. Zener diodes are available in a variety of voltages, such as 3.3, 5.1, 6.2 V and others. A zener